

Can We Reliably Detect Respiratory Diseases in Precision Farming? A Systematic Review

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Simple Summary: In this systematic review, we evaluated the use of precision livestock farming (PLF) technologies to detect respiratory diseases in livestock. This can help farmers to be able to fulfill their purpose and use PLF options. Few PLF technologies and using a reliable reference test, signs or the onset of respiratory diseases were assessed previously published devices for respiratory health, and it highlights when discussing the potential and

Abstract: Respiratory diseases compromise productivity and welfare. The use of PLF technologies for disease detection has been developed for several PLF technologies can reliably monitor respiratory health if high precision (> 90%) under field conditions and only technologies tested in studies included—swine (13), poultry (6), however, two of these were considered fully fit our criteria. Future studies to assess technology's performance in field tests.

Keywords: sensor; smart farming;

1. Introduction

Worldwide livestock respiratory diseases are increasing and death risk [1]. In swine, 3% [2] to 49.3% [3] while flock prevalence ranges from 0.5% to 80% [4]. In swine, 38.5% [5] prevalence ranges from 0.5% to 47%.

Many pathogens are responsible for respiratory diseases but the clinical manifestation is sneezing, lethargy, and nasal discharge. Precision Livestock Farming (PLF) is a promising tool for monitoring to aid with farm management.

系统综述 能否通过精准农业可靠地检测呼吸疾病？

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简述：在这项系统综述中，我们评估了精准畜牧业（PLF）技术是否能够实现其可靠参考测试时，很少有PLF技术的临床体征或呼吸道疾病的发作。本文旨在评估先前发表的用于评估呼吸健康的技术，并强调在讨论PLF技术的

摘要：呼吸道疾病通常影响牲畜的生产力和福利。在呼吸道疾病检测中的应用已为多种PLF技术。少数PLF技术和使用可靠的参考测试，临床体征或呼吸道疾病的发作。本文旨在评估先前发表的用于评估呼吸健康的技术，并强调在讨论PLF技术的

关键词：传感器；智慧农业；生物

1. 引言

全球牲畜呼吸道疾病高发，患病率范围从3% [2] 到49.3% [3]。在猪中，38.5% [5] 的患病率范围从0.5%到80% [4]。由于呼吸道疾病导致猪的死亡率高达47% [1]，牛中为19% [7,8]。

许多病原体导致牲畜患病，临床体征包括咳嗽、打喷嚏、嗜睡

精准畜牧业（PLF）定义为

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用：加里多，L.F.C.；佐藤，S.T.M.；科斯塔，L.B.；达罗斯，R.R. 能否通过精准农业可靠地检测呼吸疾病？一项系统综述。《动物学》**2023**, *13*, 1273。
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tory Diseases through Precisio

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matic review, we assessed studies on automatic monitoring of respir
n help to understand if precision livestock farming (PLF) technologie
nd can provide insights into the potential of commercially availabl
ies had good performance measures with tests under field condition
st, indicating that technologies still fall short for monitoring clinic
y diseases in bovine, swine, and poultry productions. This revie
development and validation articles of PLF technologies to monit
ghts that more than just performance measures should be assesse
nd pitfalls of PLF technologies.

commonly affect livestock species, negatively impacting animal'
se of precision livestock farming (PLF) applied in respiratory diseas
r several species. The aim of this systematic review was to evaluate
onitor clinical signs or detect cases of respiratory diseases. A techno
igh performance was achieved (sensitivity > 90% and specificity
ditions and using a reliable reference test. Risk of bias was assesse
studies with low risk of bias were considered reliable. From 23 studie
6), and bovine (4) —only three complied with our reliability criteri
sidered to have a high risk of bias. Thus, only one swine technolog
ies should include field tests and use previously validated referenc
formance. In conclusion, relying completely on PLF for monitorin
hallenge, though several technologies are promising, having hig

ng; bioacoustic; animal welfare; health monitoring

piratory diseases are highly prevalent, reducing productivit
. In poultry production, animal-level prevalence ranges fro
ck-level respiratory diseases prevalence often reaches ove
5] of animals are affected by this malady, and in cattle th
to 61% [1]. Mortality due to respiratory diseases is relativel
7% in poultry [6], 4% in swine [5], and 19% in bovine [7,8].
sponsible for causing respiratory diseases in livestock [9–11]
is similar in most cases. The main clinical signs are coughin
l and ocular discharge [9,11].
ming (PLF) is defined as automated continuous animal mo
anagement [12]. PLF technologies are based on the use

测呼吸道疾病？一项系统综述

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评估了关于牲畜呼吸道疾病自动监测的研究。这有助于理解精准畜
其目标，并可以提供对市售PLF选项潜力的见解。在田间条件下使
术具有良好的性能指标，这表明技术仍无法监测牛、猪和家禽生产
。本综述评估了先前发表的用于监测呼吸道健康的PLF技术开发和
的潜力和陷阱时，除了性能指标外，还应评估更多因素。

品种，对动物的生产力和福利产生负面影响。精准畜牧业（PLF）
多种物种开发。本系统综述的目的是评估PLF技术是否能够可靠地
例。如果一项技术在田间条件下使用可靠参考测试时达到高灵敏度
，则被认为可靠。评估了偏倚风险，仅考虑了在低偏倚风险研究中
研究（猪13项、家禽6项、牛4项）中——仅三项符合我们的可靠性
有高偏倚风险。因此，只有一项猪技术完全符合我们的标准。未来
先前验证的参考测试来评估技术的性能。总之，完全依赖PLF监测
几种技术很有前景，在田间测试中表现出高性能。

物声学；动物福利；健康监测

，导致生产力下降并增加死亡风险 [1]。在家禽生产中，动
2] 到 49.3% [3]，而群水平的呼吸道疾病患病率通常超过
5] 的动物受此疾病影响，而在牛中，患病率范围从 0.5% 到
致的死亡率在牲畜中相对较高，家禽中为 47% [6]，4%，猪中

呼吸道疾病 [9–11]，但大多数情况下临床症状相似。主要
睡以及鼻部和眼部分泌物 [9,11]。

自动化持续动物监控以辅助农场管理 [12]。PLF技术基于使用

sensors to monitor animals and dairy calves [13], poultry [14], a diseases by monitoring signs, types of diseases such as New nologies can detect the presenc individual level [16].

To assess if a technology h commercial farms), several poin validation conditions must be s

Performance measures incl itive predictive value (PPV), an and Kristensen [18], if the perf further assess the time window ultimately arise from the result is used as a reference to assess objectives [19]. Thus, when eval of the chosen reference test.

The study condition may situation (i.e., commercial setti with field condition tests is like

Performance measures ma type of sensors applied. For exa to nonspecific signs of disease, detecting a specific type of dise the total number of cases (i.e., e prediction) are biased by the tru rates compared to diseases that all this information with perfor

There have been recent sy However, these reviews focuse once (e.g., lameness, respiratory devices developed exclusively f the availability of information effectiveness of a technology is To support that the technology the study should present the stu the technology was tested.

The objectives of this syst monitoring respiratory diseases (2) the quality of studies in re possible to rely on PLF to auto

2. Materials and Methods

2.1. Literature Search

A systematic literature rev nologies for the detection of res

The PRISMA guideline [24] was conducted in the Scopus, The search field in Web of Scie keywords". Search terms are pr

These terms were sorted dition" terms. The search stri Boolean operator "AND" and " sion Livestock Farming" OR "N

传感器来监测动物，并已开发用过监测咳嗽、打喷嚏和发烧等体疫和猪消瘦病。这些技术可以在

为评估一项技术是否具有外几点：高性能指标，且验证条件

性能指标包括特异性（Sp和阴性预测值（NPV）。根据D人满意，则无需进一步评估时间金标准）的结果，该测试被用作估精准畜牧业（PLF）开发时，

研究条件可能决定性能是否条件测试验证的技术效率很可能

性能指标也可能受到所收集疾病非特异性症状相关数据（如外，考虑病例总数（即，每项真疾病患病率的偏差影响，罕见疾息与性能指标进行严格结合，才

关于精准畜牧业在疾病监测注的是同时针对多种福利相关问地研究专门用于监测呼吸健康状况理解一项技术的有效性远比仅看好，研究应说明研究条件、参考

本系统综述的目的是评估：吸系统疾病的进展情况，(2) 研精准畜牧业自动监测牲畜种类的

2. 材料与方法

2.1. 文献检索

进行了一项系统文献综述，的数据。

PRISMA指南 [24] 被用于 Web of Science和IEEE数据库 Scopus为“文章标题、摘要、

这些术语被分为“物种”术布尔运算符“AND”和“OR” AND “精准畜牧业” OR “非

nd have been developed for monitoring respiratory health i], and swine [15]. These technologies can monitor respirator s, such as cough, sneezing, and fever or detecting specifi ewcastle disease and porcine wasting diseases. These tec ence of diseases or clinical signs at the group level [15,16] o

y has external validity (i.e., is effective to be implemented i oints should be observed: high performance measures and th e similar to the conditions in commercial farms [17].

nclude specificity (Sp), sensitivity (Se), precision, accuracy, po and negative predictive value (NPV). According to Dominia erformance measures are not satisfactory there is no need t w and validation conditions. The performance measurement ults of a reference test (usually called a gold standard), whic ss if the developed technology was efficient in achieving it valuating PLF development, it is critical to assess the adequac

ay determine if the performance is replicable in a real-lif etting); therefore, the efficiency from technologies validate ically to be similar to the efficiency on commercial farms [18] may also be influenced by the type of data collected and th xample, sensors that are only capable of recording data relate se, such as elevated body temperature, are not indicated fo isease [20]. In addition, performance measures that consid e., error rate, number of false positives for every true positiv rue disease prevalence, with rarer diseases having higher erro hat are highly prevalent [18]. Therefore, critically combinin orformance measures allows one to assess technology readines systematic reviews about PLF on disease monitoring [20–23 sed on technologies that target many welfare-related issues ory diseases, diarrhea). A more profound investigation of PL y for monitoring respiratory health status is needed to enhanc n on the efficiency of these technologies. Understanding th is much more complex than only looking at its performanc gy is reliable and able to perform well in real-life condition study's condition, reference test, and a full description of ho

ystematic review were to assess: (1) the advances of PLF fo ses at group or animal level in different livestock production eporting the effectiveness of the technology, and (3) if it i tomatically monitor respiratory diseases in livestock species

review was conducted to gather data on validated PLF tec respiratory diseases in livestock species.

[24] was used for the systematic literature search. The searc s, Web of Science, and IEEE databases on 21 September 202 cience was “topic”, and in Scopus was “article title, abstrac presented in Table 1.

d as “species” terms, “technology” terms and “type of co tring consisted of a combination of terms separated by th d “OR”; for example: “Dairy Cow*” OR “Cow” AND “Prec “Non-invasive Technology” AND “Respiratory Disease*” O

用于监测奶牛小牛 [13], 家禽 [14], 和猪 [15]。这些技术可以 体征来监测呼吸系统疾病, 或检测特定类型的疾病, 如新城 在群体水平或个体水平检测疾病或临床体征或个体水平 [16]

外部效度 (即, 是否能在商业农场有效实施), 应观察以下 件必须与商业农场条件相似 [17]。

p)、灵敏度 (Se)、精确度、准确度、阳性预测值 (PPV Dominiak和Kristensen [18], 的研究, 如果性能指标不令 间窗口和验证条件。性能测量最终源于参考测试 (通常称为 作评估所开发技术是否有效实现其目标的参考。因此, 在评 , 必须评估所选参考测试的充分性。

否能在实际场景 (即, 商业环境) 中复制; 因此, 通过田间 能与商业农场的效率相似 [18]。

集数据类型和所应用传感器类型的影响。例如, 仅能记录与 如体温升高) 的传感器不适合检测特定类型的疾病 [20]。此 真阳性预测的误报率、假阳性数量) 的性能指标会受到真实 疾病的误报率高于高患病率疾病 [18]。因此, 将所有这些信 才能评估技术的成熟度。

测方面的系统综述已有近期发布 [20–23]。然而, 这些综述 问题 (如跛行、呼吸系统疾病、腹泻) 的技术。需要更深入 状况的精准畜牧业设备, 以提升这些技术效率信息的可用性 看其性能复杂。为证明该技术可靠且能在实际条件下表现良 考测试, 并全面描述该技术是如何被测试的。

: (1) 精准畜牧业在不同畜牧业生产中用于监测群体或个体 研究在报告该技术有效性方面的质量, 以及 (3) 是否可以依 的呼吸道疾病。

, 以收集关于牲畜种类中呼吸道疾病检测的验证精准畜牧业技术

于系统文献检索。该检索于2022年9月21日在Scopus、 库中进行。Web of Science的检索字段为“主题”, 、关键词”。检索词见表 1。

术语、“技术”术语和“疾病类型”术语。搜索字符串由用 ”分隔的术语组合而成; 例如: “奶牛*” OR “奶牛” 非侵入性技术” AND “呼吸系统疾病*” OR

“Cough”. To filter the search, in and “Conference paper” was se

Table 1. Search strings for the liter

Species Terms

Dairy cow(s)
Cow
Cattle
Calf
Calves
Pig
Sow
Swine
Broiler
Laying hen
Chicken
Poultry
Goat
Sheep
Ewe
Lamb

The inclusion criteria wer the development or validation production animals (this inclu technologies developed for dise not develop, validate, or apply or respiratory diseases clinical s other disease were not include or improvement of a specific te event or differentiation betwee developed for other species that

Two researchers were invol and S.T.M. Sato). The researchers databases. The inclusion and exc researcher. The results of the scre exclusion of articles. This step w The screening started with the e related to PLF were excluded b the inclusion and exclusion crite remaining studies were include Conflicted studies were analyzed review after agreement if the stud about the number of excluded a presented in Figure 1.

Three articles were exclu article was just available in Ma inaccurate translation.

All articles included in t technologies to monitor respira

“咳嗽”。为了筛选搜索，在W了“文章”和“会议论文”。

表1。文献检索的搜索字符串。

物种术语

奶牛（只）
Cow
牛
Calf
小牛
Pig
Sow
猪
肉鸡
产蛋鸡
鸡
家禽
Goat
绵羊
Ewe
Lamb

纳入标准为在任何时间发表业（PLF）技术的原始研究（这排除标准为未开发、验证或应用征的研究（即，关于一般健康监特定技术功能的研究文章，例如“物种”术语中未提及的其他物

两位研究人员参与了初始的人员使用相同的术语和数据库独行。检索后，比较了筛选过程的从排除重复研究开始。其次，通析标题、摘要和关键词后，根据余的研究被纳入或排除。有争议合纳入或排除标准）后纳入或排图1。

由于无法找到全文，有3篇译不准确导致错误而被排除。

本次综述中所有纳入的文章

, in Web of Science and Scopus platform the option “Article s selected in “Document Types”.

terature search.

Technology Terms	Type of Conditions Terms
Precision Livestock Farming	Respiratory Disease(s)
Noninvasive Technology	Cough
Smart sensor	Fever
Smart Farming	BRD
Automated technology	Vocalization
Online health monitoring	Infectious disease(s)
Computer vision	Sneeze
Cough recognition	Respiratory disease detection
Sound analysis	
Sound classification	
Convolutional neural network	

ere original research studies published at any time abo on of PLF technologies for monitoring respiratory disease i cluded technologies aimed at monitoring clinical signs an isease detection). The exclusion criteria were studies that di ly PLF technologies to exclusively detect respiratory disease al signs (i.e., research papers on general health monitoring o ded); this included research articles about the developme technology’s feature, for example, localization of the coug een types of coughs were excluded. Studies of technologie hat are not present in the “species” terms, were also exclude olved in the initial search and screening process (L.F.C. Garrid rs worked independently on the search using the same terms an exclusion of articles were also conducted independently by eac screening process were compared after the search and inclusion was conducted to ensure that all relevant studies were include e exclusion of duplicate studies. Secondly, studies that were n d by title and abstract. Then studies were screened followin iteria after analyzing the title, abstract, and keywords [25]. Th ded or excluded after analyzing the entire content of the articl ed together by screeners and were included or excluded from th udy complied with the inclusion or exclusion criteria. Informatio d and included studies and the reason for excluding studies i

cluded because we were unable to find the full text. On Mandarin and therefore excluded to prevent mistakes due t

the review were used for assessing the advances of PL ratory diseases.

Web of Science和Scopus平台上，在“文献类型”中选择

技术术语	疾病类型术语
精准畜牧业	呼吸系统疾病（种）
非侵入性技术	咳嗽
智能传感器	发烧
智慧农业	BRD
自动化技术	发声
在线健康监测	传染病
计算机视觉	打喷嚏
咳嗽识别	呼吸道疾病检测
声音分析	
声音分类	
卷积神经网络	

表的关于为监测生产动物呼吸道疾病而开发或验证精准畜牧 这包括旨在监测临床体征的技术和为疾病检测开发的技术) 用精准畜牧业技术专门检测呼吸道疾病或呼吸道疾病临床体 监测或其他疾病的研究未被纳入)；这包括关于开发或改进 如，咳嗽事件的定位或咳嗽类型之间的区分被排除。对于 物种而开发的技术的研究也被排除。

的检索和筛选过程 (L.F.C. Garrido和S.T.M. Sato)。研 独立进行检索。文章的纳入和排除也由每位研究人员独立进 的成果。这一步骤是为了确保所有相关研究都被纳入。筛选 通过标题和摘要排除与精准畜牧业无关的研究。然后，在分 据纳入和排除标准进行筛选。[25]。在分析文章全文后，剩 议的研究由筛选人员共同分析，并在达成一致（如果研究符 排除于综述。排除和纳入研究的数量以及排除研究的原因在

篇文章被排除。其中1篇文章仅提供中文版本，因此为避免翻

章均用于评估精准畜牧业技术监测呼吸道疾病的进展。

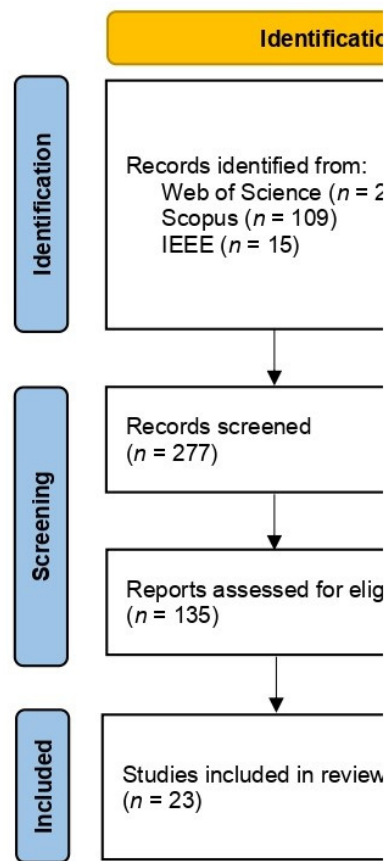


Figure 1. Modified Preferred Repo flow diagram [24] containing information on the selection process. The PRISMA checklist is available in the supplementary materials.

2.2. Data Gathered from Articles

After the screening process, the data gathered from the articles was categorized by species, the technology was applied (laboratory or field condition), the study was performed in a university or not, the environment was similar to a commercial farm or not, the type of sensor/monitoring system used (if found in the literature), the performance measure presented (e.g., feed conversion ratio, FCR), and the reference test used (e.g., standard error of the mean, SEM).

The animal production stage was determined based on the species. For swine production, we determined the stage when the articles did not clarify the age of the animals when they were ten to sixteen weeks old as the finishing stage. For bovines, the stage was determined when the animals were older than sixteen weeks as the finishing stage. For poultry, the age was determined.

Information regarding the type of production system (broiler, dairy, or meat production) and the experimental setup (laboratory or field) was also collected.

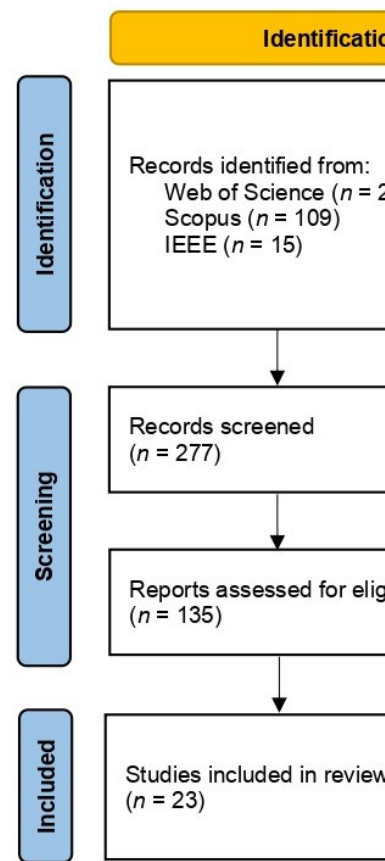


图1. 改良后的系统评价与荟萃分析选择信息的流程图。PRISMA清单可在补充材料中找到。

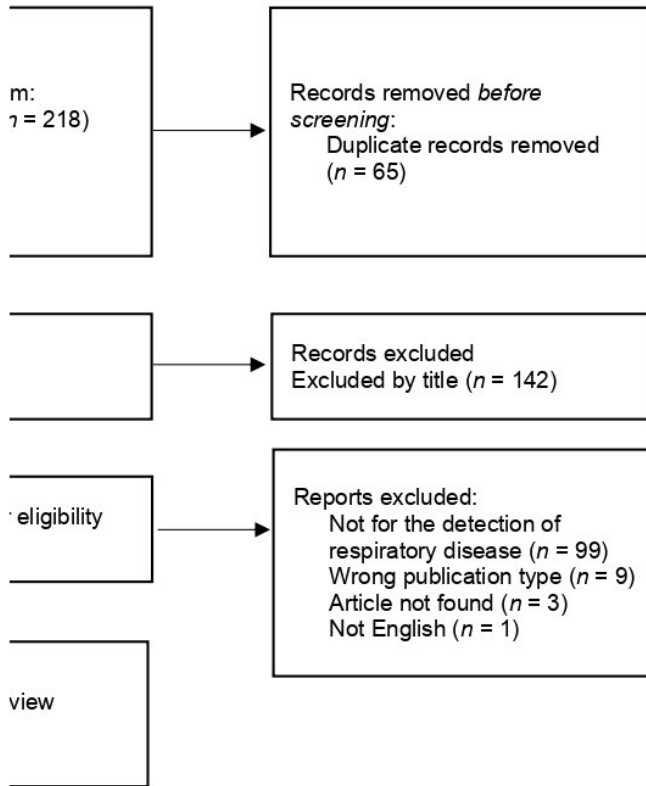
2.2. 文章收集的数据

筛选过程后，从纳入文章中收集的数据按物种、技术类型（实验室条件或田间条件：当研究在受控环境中进行时，被认为是在受控环境中模拟田间条件）、研究是否在大学进行、环境是否与商业农场相似、传感器/监测系统类型（如果在文献中发现的任何其他传感器）、呈现的任何其他性能指标（例如，饲料转化率，FCR）和参考测试（例如，标准误差，SEM）进行分类。

根据物种确定生产阶段。对于生猪生产，当文章未明确动物年龄时，将10至16周龄定为育肥阶段。对于牛，将16周龄以上定为育肥阶段。对于家禽，根据年龄确定阶段。

也会收集。对于在实验室条件下进行的研究，也会收集。

Identification of studies via databases and registers



porting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the strategy for the systematic review search and study selection is presented as Supplementary Table S1.

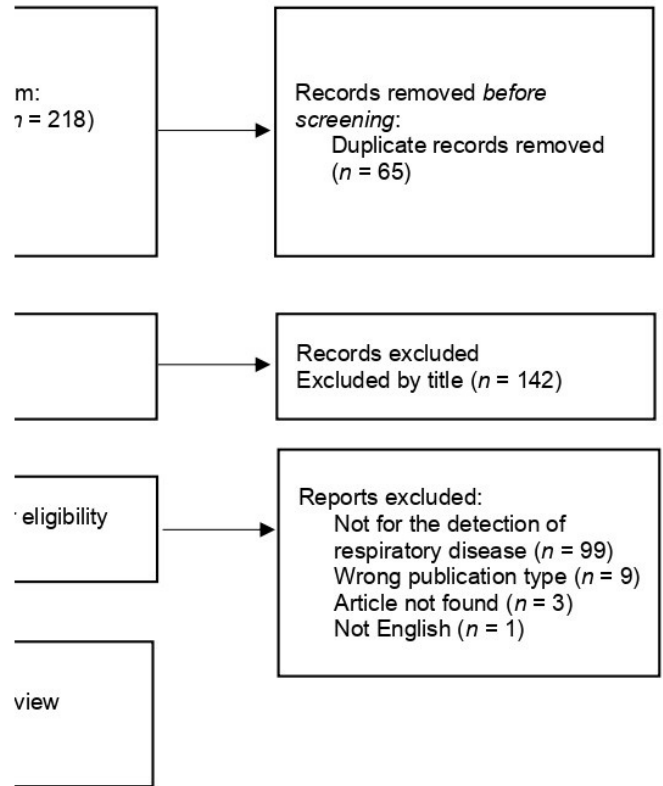
es

ess, the following data were extracted from included article: the technology used, the conditions in which the study was conducted (laboratory conditions or field conditions), the research center or conducted in a controlled environment, the type of animal used (sound-based, image-based, or any other sensor), the performance measures (Se, Sp, precision, accuracy, PPV, PNV), the results presented by the study, if recall was presented we considered it, and the technology used for validating the technology.

stage in which these technologies are tested was also assessed. The production stage was determined based on animals' age or weight. Animals were considered in the fattening stage if they were less than 16 weeks old and weighed between 30 and 60 kg. Animals that weighed more than 60 kg were considered in the fattening stage. The stage was considered the type of animal used (e.g., calves).

The production system and the aim of the production (e.g., egg production) was also gathered when presented by the article. For laboratory conditions, the production system was considered as a

Identification of studies via databases and registers



分析报告规范（PRISMA）流程图 [24] 包含系统评价检索策略和研究清单作为补充表S1呈现。

中提取了以下数据：技术应用的物种、研究进行的环境条件（研究在大学或研究中心进行或在一个与商业农场不相似的受控实验室条件下进行的；田间条件被认为是在商业农场进行时）、使用的传感器/设备类型（基于声音的、基于图像的或基于性能的）、性能指标（Se、Sp、精确度、准确度、PPV、PNV或研究如果呈现了召回率，则将其视为Se），以及用于验证技术的

阶段也进行了评估。对于猪生产，当文章未明确说明阶段时，默认生产阶段。当动物年龄为十至十六周且体重在30至60公斤，年龄超过十六周且体重超过60公斤的动物被视为出栏阶段。对于牛（例如，小牛、公牛）确定，而家禽则根据年龄确定。

（例如蛋、肉鸡、奶牛或肉的生产）的信息，在文章提供时，生产系统被视为一个实验设置。

2.3. Risk of Bias

We assessed the risk of bias to understand the consistency considered to have a low risk presented. Similar to Stygar et al., the article had to provide the following information: the study was held (laboratory or field), the device used (manufacturer of the device), the software used (version), the data used (raw data or processed data). In addition, we added the assessment of bias; namely, the population of animals and raw data used to have a high risk of bias if on

2.4. Technology Reliability

In this review, the word “reliable” refers to respiratory diseases or to monitoring, and the reference test used is reliable. Only studies with a low

For dairy cows, the International Standard Organization presents a minimum performance (Se > 90% and Sp > 99%). These minimum values are those that farmers have identified as a performance above this threshold of PLF technologies for monitoring. The rationale is that of the ISO 2096 for monitoring respiratory diseases. We assessed the validated technology (Se > 90% and Sp > 99%) or precision was the technology presented. If Sp and precision had to be higher than 90% and 99%, they were not included in the discussion. For methods for developing the technology (used), we collected the performance.

The reference test was considered effective to assess the clinical situation. If the test reliability, tests should be able to detect the disease or clinical use of a well-established reference for it. For sound-based recordings, we considered the labeling process [29].

In sum, the criteria for conducting a reliable reference test, tests with a low risk of bias.

3. Results

A total of 23 articles were included in the review. Technologies for swine (13/23), and for cattle (10/23). No articles were found about P

Most technologies used so far were image-based (1/23) and data-based (1/23) from the studies included in this

2.3. 偏倚风险

我们评估了本次综述所纳入的研究的偏倚风险 [26]。当所有可能影响结果的因素都得到控制时，Stygar等人 [23] 和Hendriks等人 [24] 的研究被认为是低偏倚风险：研究进行的环境（实验室（所用传感器的制造商、距离传感器的位置）或使用的机器学习方法）。此外，研究描述（年龄、体重或生产阶段）或多项关键信息未得到呈现，研

2.4. 技术可靠性

在本综述中，“可靠”一词是指通过评估性能、研究条件偏倚风险低的研究的可靠性。

对于奶牛，国际标准化组织提出了性能要求（Se > 80%和Sp > 90%）。这些最低要求是农民已经识别出的性能高于这个阈值的PLF技术监测呼吸疾病的成功率为80%以上。我们评估了验证过的技术（Se > 90%和Sp > 99%）或精度是否达到了所提出的技术中，Se和Sp都必须高于90%。当一项研究比较了用于开发技术时，我们收集了性能最高的技术的性能。

参考测试被认为可靠，当研究明确参考测试的可靠性，文章应展示一种考虑可靠性的方式是使用已验证的技术。对于基于声音的技术，由于标注声音不被认为是可靠的 [29]。

总之，考虑一项技术可靠的进行测试，以及研究具有低偏倚

3. 结果

共有23篇文章被纳入本次综述。（6/23）和牛生产（4/23）。没有发

大多数技术使用基于声音的（1/23）和加速度计（1/23）。

bias in studies included in this review. This process was helped by the results based on the study design [26]. A study is at risk of bias when all information that could bias the results is not reported. In the study by Hendriks et al. [27], we considered that the following information to prevent bias: conditions in which the study was conducted (indoor or field conditions), housing, type of sensor, and how it was used (what type of software, what machine learning method was used). We considered three more pieces of information that had to be presented to reduce the risk of bias: location description (age, weight or production stage), number of animals used to calculate performance measures. A study was categorized as having a low risk of bias if one or more essential information was not presented.

“reliable” refers to the sole use of PLF technologies to detect the clinical signs of disease. Performance, study conditions, and the risk of bias were assessed to decide if a technology was considered to have a low risk of bias. The International Organization for Standardization (ISO 20966, 2007) [2] defines the minimum performance required to support a technology efficiency (Se > 80%). If the minimum requirements are supported because the reported success in clinical mastitis is 80%, and therefore a technology with a success rate of 80% is able to outperform the human [18]. To assess performance in monitoring respiratory diseases, it would be ideal to use a similar standard as in ISO 20966, though to our knowledge there are no specific standard for diseases in livestock. Therefore, similarly to Stygar et al. [23], we assessed the technology's performance by considering high performance when the success rate was >90%. Thus, both Se and Sp had to be higher than 90% for a technology to be considered reliable. If the success rate was not presented but the article presented precision, both Se and Sp had to be higher than 90%. Articles that did not present Se and Sp or precision were considered not reliable technologies. When a study compared various technologies (e.g., when multiple machine learning methods were used), the performance of the technology that achieved the highest performance was considered reliable when the study presents evidence that it is able to detect the clinical sign that is being monitored. For an article to prove the reference test was reliable, the study presented demonstrating that the used reference test was able to detect the clinical sign. Another way of considering reliability was the reference test, given that the authors provided peer-reviewed technologies, remote audio labeling sounds from audiotapes, and the study was considered reliable due to misclassifications that may occur in the field.

Studies considered a technology reliable were high performance, used reference tests conducted in field conditions, and the study having low risk of bias.

Studies included in this review. Most of the articles were on PLF technologies (13/23), followed by poultry (6/23) and bovine production (4/23). The most common PLF technologies for small ruminants (goats and sheep) were sound-based devices (21/23). Other types of sensors found were accelerometers (1/23). Table 2 summarizes the data gathered in this review.

Studies included in this review. This process was helped by the results based on the study design [26]. A study is at risk of bias when all information that could bias the results is not reported. In the study by Hendriks et al. [27], we considered that the following information to prevent bias: conditions in which the study was conducted (indoor or field conditions), housing, type of sensor, and how it was used (what type of software, what machine learning method was used). We considered three more pieces of information that had to be presented to reduce the risk of bias: location description (age, weight or production stage), number of animals used to calculate performance measures. A study was categorized as having a low risk of bias if one or more essential information was not presented.

词指的是仅使用PLF技术来检测呼吸道疾病或监测疾病临床条件和参考检测方法，以确定技术是否被认为可靠。仅评估了

织 (ISO 20966, 2007) [28]提出了支持技术效率所需的最低性能要求 (>99%)。这些最低要求之所以得到支持，是因为农民报告的成功率在80%，因此性能高于此阈值的技术能够优于人类 [18]。为了评估性能，理想的是使用与ISO 20966类似的推理，尽管根据我们研究的特定标准。因此，类似于Stygar等人 [23]，Se或精确度是 >90%时性能较高，来评估验证技术的性能。Sp都必须高于90%。如果未提供Sp，但文章提供了精确度，未提供Se和Sp或精确度的文章未包含在可靠技术的讨论中。各种方法（例如，当使用了多种机器学习方法时），我们

研究提供证据表明其能有效评估所监测的临床体征时。要证明展示所用参考测试能够检测疾病或临床体征的测试结果。另已建立的参考测试，前提是作者提供了同行评审的参考文献。注过程中可能出现的错误分类，音频录音中的远程音频标注

的标准包括：高性能、使用可靠的参考测试、在领域条件下低偏倚风险。

。其中大多数文章是关于猪的精准畜牧业技术 (13/23)，其次是家禽发现关于小反刍动物（山羊和绵羊）的精准畜牧业技术的文章。

的设备 (21/23)。其他发现的传感器类型包括基于图像的设备。表 2 总结了本次综述所纳入研究收集的数据。

Table 2. Information about sensor type of included articles for swine, poultry

Species	Study	Sensor Type	Performance Measures
Swine	[30]	Sound Based	Positive cough recognition
	[31]	Sound Based	Positive cough recognition
	[32]	Sound Based	Accuracy
	[33]	Sound Based	Accuracy
	[34]	Sound Based	Correct identification ratio
	[35]	Sound Based	Correct identification ratio
	[36]	Sound Based	Accuracy
	[37]	Sound Based	Sensitivity, Precision, Accuracy, and cough detection rate, and word error rate
	[15]	Sound Based	Sensitivity, Precision, Accuracy, and word error rate
	[38]	Sound Based	Sensitivity, Specificity, Precision, Accuracy, and F1-score
	[39]	Sound Based	Sensitivity, Specificity, Precision, Accuracy, and F1-score
Poultry	[40]	Sound Based	Sensitivity, Precision, Accuracy, and F1-score
	[41]	Sound Based	Sensitivity, Precision, Accuracy, and F1-score
	[42]	Sound Based	Sensitivity, Specificity, and Accuracy
	[43]	Sound Based	Sensitivity, Precision, and Accuracy
	[14]	Sound Based	Sensitivity, Specificity, and Precision
Bovine	[44]	Sound Based	Accuracy
	[45]	Sound Based	Sensitivity, Precision, Accuracy, and F1-score
	[46]	Sound Based	Sensitivity, Precision, Accuracy, and F1-score
	[47]	Image	Sensitivity, Specificity, PPV, NPV, and Cut-off
	[13]	Sound Based	Sensitivity, Specificity, and Precision
	[48]	Sound Based	Sensitivity, Specificity, and Precision
	[16]	Accelerometer	Sensitivity, Specificity, Accuracy, and M

The production stage, production are presented in the Supplemental Figure 2, accessed <https://doi.org/10.6084/m9.figshare.21758543.v2>, accessed

3.1. Studies' Condition and Refere

In this section, we report a The reference test varied in relation tests found were: blood analysis (2/23; assessment of clinical significance analysis (1/23; number of neutrophils (11/23; remote audio labeling (11/23; (3/23), PCR (2/23), remote audio collected blood samples to identify used both image of the animals blood samples to identify the d

3.2. Risk of Bias

A total of eight studies were included in the meta-analysis. Two production studies [32,33,37,38,41], two production study [47]. Table 3 presented in each study.

表2。关于猪、家禽和牛生产的传感器类型

物种研究	传感器类型	性能指标	
猪	[30]	基于声音	阳性咳嗽识别
	[31]	基于声音	阳性咳嗽识别
	[32]	基于声音	准确度
	[33]	基于声音	准确度
	[34]	基于声音	正确识别率
	[35]	基于声音	正确识别率
	[36]	基于声音	准确度
	[37]	基于声音	灵敏度、精确度、准确度以及咳嗽检测率
	[15]	基于声音	灵敏度、精确度、咳嗽检测率和F1分数
	[38]	基于声音	词错误率
	[39]	基于声音	灵敏度、特异性、精确度、准确度和F1分数
家禽	[40]	基于声音	灵敏度、特异性、精确度、准确度和F1分数
	[41]	基于声音	灵敏度、精确度、准确度和F1分数
	[42]	基于声音	灵敏度、特异性和准确度
	[43]	基于声音	灵敏度、精确度、准确度
	[14]	基于声音	灵敏度、特异性、精确度
	[44]	基于声音	准确度
	[45]	基于声音	灵敏度、精确度、准确度和F1分数
牛	[46]	基于声音	灵敏度、精确度、准确度和F1分数
	[47]	图像	灵敏度、特异性、阳性预测值、阴性预测值和
	[13]	基于声音	灵敏度、特异性和精确度
	[48]	基于声音	灵敏度、特异性和精确度
	[16]	加速度计	灵敏度、特异性、准确度和MCC

所有物种的生产阶段、生产 [https://doi.org/10.6084/m9 .fi](https://doi.org/10.6084/m9.figshare.21758543.v2)

3.1. 研究条件与参考测试

在本节中，我们报告了本次和技术应用的不同而有所差异。数量）、临床评估（2/23；评估性粒细胞数量和针对呼吸系统疾病中的声音）、实时音频标注（23；标注声音并收集血液样本和音频录音标注咳嗽声音并收集

3.2. 偏倚风险

共有八项研究被认为存在高风险生产研究 [43,44]，一项是牛生产

or type, performance measures, study conditions, and reference to poultry, and bovine productions.

	Study Conditions	Reference Test
detection rate , and F1-score , and F1-score , and F1-score F1-score	Laboratory	Remote audio labeling
	Laboratory	Remote audio labeling
	Field	Live audio labeling
	Field	Live audio labeling
	Laboratory	Remote audio labeling
	Field	Remote audio labeling
	Field	Live audio labeling
	Field	Remote audio labeling and blood analy
	Field	Video labeling and blood analysis
	Laboratory	Remote audio labeling
racy acy sion	Field	Remote audio labeling
	Field	Remote audio labeling
	Field	Remote audio labeling
	Field	Remote audio labeling
	Field	Remote audio labeling
F1-score F1-score	Laboratory	PCR
	Laboratory	Remote audio labeling
	Laboratory	Remote audio labeling
ut off value sion sion d MCC	Laboratory	PCR
	Field	Remote audio labeling
	Laboratory	Video labeling and PCR
	Field	Clinical assessment
	Field	Clinical assessment and blood analysi
	Field	Blood analysis
	Field	Clinical assessment

production system, and aim of the production of all specie mentary Table S2 available at [[https://doi.org/10.6084/m](https://doi.org/10.6084/m ed on 14 March 2023) ed on 14 March 2023].

ference Tests

rt all reference tests found in studies included in this revie relation to the species and the technology applied. Referenc sis (1/23; used the number of neutrophils), clinical assessme signs presented by the animals), clinical assessment and bloo utrophils and clinically assessment for respiratory diseases ; manually label sounds from audio files), live audio labelin audio labeling and blood analysis (1/23; label sounds an identify the disease), video labeling and blood analysis (1/2 als and audio recordings to label cough sounds and collecte e disease), video labeling, and PCR (1/23).

were considered to have a high risk of bias: five swine produ two poultry production studies [43,44], and one for bovin 3 shows the specific information that was presented or n

传感器类型、性能指标、研究条件和参考测试的包括文章信息。

	研究条件	参考测试
	实验室	远程音频标注
	实验室	远程音频标注
	领域	实时音频标注
	领域	实时音频标注
	实验室	远程音频标注
	领域	远程音频标注
	领域	实时音频标注
	领域	远程音频标注和血液分析
	领域	视频标注和血液分析
	实验室	远程音频标注
	领域	远程音频标注
	领域	远程音频标注
	领域	远程音频标注
	领域	远程音频标注
	领域	远程音频标注
	实验室	PCR
	实验室	远程音频标注
	实验室	远程音频标注
	实验室	PCR
	实验室	远程音频标注
	实验室	远程音频标注
和截断值	实验室	视频标注和PCR
	领域	临床评估
	领域	临床评估和血液分析
	领域	血液分析
	领域	临床评估

产系统及生产目的均呈现在补充表S2中，该表可在 [figshare.21758543.v2获取，数据访问日期为14 March 2023]

次综述所纳入研究中使用的所有参考测试。参考测试因物种 。发现的参考测试包括：血液分析（1/23；使用中性粒细胞 估动物呈现的临床体征）、临床评估和血液分析（1/23；中 疾病的临床评估）、远程音频标注（11/23；手动标注音频 （3/23）、PCR（2/23）、远程音频标注和血液分析（1/ 以识别疾病）、视频标注和血液分析（1/23；使用动物图像 集血液样本以识别疾病）、视频标注和PCR（1/23）。

高偏倚风险：五项是猪生产研究 [32,33,37,38,41], 两项是家 产研究 [47]。表3 展示了每项研究中呈现或未呈现的具体信

Table 3. Evaluation of the descriptive risk of bias.

Species	Study	Study Conditions	Housing	Hardware	How It Was Installed
Swine	[30]	✓	✓	✓	✓
	[31]	✓	✓	✓	✓
	[32]	✓	✓	✗	✓
	[33]	✓	✓	✗	✓
	[34]	✓	✓	✓	✓
	[35]	✓	✓	✓	✓
	[36]	✓	✓	✓	✓
	[37]	✓	✓	✓	✓
	[15]	✓	✓	✓	✓
	[38]	✓	✗	✓	✗
	[39]	✓	✓	✓	✓
Poultry	[42]	✓	✓	✓	✓
	[43]	✓	✓	✗	✓
	[14]	✓	✓	✓	✓
	[44]	✓	✓	✓	✓
	[45]	✓	✓	✓	✓
Bovine	[47]	✓	✓	✓	✓
	[13]	✓	✓	✓	✓
	[48]	✓	✓	✓	✓
	[16]	✓	✓	✓	✓

✓—Information provided in the article.

3.3. Respiratory Disease PLF Tech

All the technologies found depending on the methodolog vocalization (3/6), sneeze (1/6 technologies for poultry produ

Table 4. PLF performance measure

	Sensitivity (%)
[42]	93.30
[43]	85.20
[14]	66.70
[44]	N/A
[45]	94.10
[46]	96.60

N/A—Not Available.

From the studies included precision, accuracy, and F1-scor Banakar et al. [42] and C detect and diagnose different t were developed based on vocal conditions and used PCR tests for the detection of Avian Infl Sp, and precision). Banakar et al Avian Influenza but failed to ac Cuan et al. [46] achieved high p technology aimed at detecting All other technologies w disease. Rizwan et al. [43] mo but was unable to achieve hi technology that detects sneeze

表3. 评估研究中对八项关键信息

物种	研究	研究条件	住房	硬件	如何安装
猪	[30]	✓	✓	✓	✓
	[31]	✓	✓	✓	✓
	[32]	✓	✓	✗	✓
	[33]	✓	✓	✗	✓
	[34]	✓	✓	✓	✓
	[35]	✓	✓	✓	✓
	[36]	✓	✓	✓	✓
	[37]	✓	✓	✓	✓
	[15]	✓	✓	✓	✓
	[38]	✓	✗	✓	✗
	[39]	✓	✓	✓	✓
家禽	[42]	✓	✓	✓	✓
	[43]	✓	✓	✗	✓
	[14]	✓	✓	✓	✓
	[44]	✓	✓	✓	✓
	[45]	✓	✓	✓	✓
牛	[47]	✓	✓	✓	✓
	[13]	✓	✓	✓	✓
	[48]	✓	✓	✓	✓
	[16]	✓	✓	✓	✓

✓—文章中提供的信息。✗—文章中未提

3.3. 家禽生产呼吸系统疾病PLF

所有在家禽生产中发现的 t 同，监测到的声音可能是咳嗽和 家禽生产 technologies 的性能

表4. 家禽生产的PLF性能指标。

	灵敏度 (%)
[42]	93.30
[43]	85.20
[14]	66.70
[44]	N/A
[45]	94.10
[46]	96.60

N/A—不可用。

从纳入的研究中，刘等人 [45 确度和F1分数)。 Banakar等人 [42] 和Cua 统疾病的技术。这些技术是在实 试作为参考测试。Cuan等人 [4 指标（灵敏度、特异度和精确度 （灵敏度和特异度），但在支气 在所有提供的性能指标中均取得 其他所有技术都基于特定的 达到了高准确度（97.6%），但 [14] 开发了一种检测打喷嚏声

ptions of eight essential pieces of information in the studies to asses

Software	Population Description	Number of Animals	Raw Data	Risk of Bias
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✗	✓	high
✓	✓	✓	✓	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✗	✓	✗	high
✓	✗	✓	✓	low
✓	✓	✓	✓	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✗	high
✓	✓	✓	✓	low
✓	✓	✓	✓	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✗	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✗	✓	✓	✓	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✓	low

icle. ✗—Information not provided in the article.

echнологies for Poultry Production

nd for poultry production were sound-based (6/6). However, the monitored sounds could be cough and snore (1/6 /6), or rale sounds (1/6). The performance measures of th duction are presented in Table 4.

ures for poultry production.

Specificity (%)	Precision (%)	Accuracy (%)	F1-Score (%)
96.73	N/A	91.15	N/A
N/A	86.60	97.60	N/A
N/A	88.40	N/A	N/A
N/A	N/A	97.00	N/A
N/A	94.40	93.80	94.20
N/A	96.54	98.50	97.33

ed, Liu et al. [45] achieved good performance measures (S core) and was developed in field conditions.

Cuan et al. [44,46] developed technologies that aimed t t types of respiratory diseases in poultry. The technologie calization before and after virus inoculation under laborator ts as reference test. Cuan et al. [44] achieved a high accurac fluenza but did not present other performance measures (S t al. [42] achieved a high performance (Se and Sp) for detectin achieve a high Se for Bronchitis Virus and Newcastle Diseas h performance for all performance measures presented, with g Newcastle disease.

were based on specific sounds for monitoring respirator monitored rale sounds and achieved high accuracy (97.6 high Se and precision. Carpentier et al. [14] developed ze sounds; however, the technology Se was low and therefor

息 的描述以评估偏倚风险。

软件	人口描述	动物数量	原始数据	偏倚风险
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✗	✓	high
✓	✓	✓	✓	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✗	✓	✗	high
✓	✗	✓	✓	low
✓	✓	✓	✓	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✗	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✗	✓	✓	✓	high
✓	✓	✓	✓	low
✓	✓	✓	✓	low
✓	✓	✓	✓	low

未提供的信息。

LF技术

technologies 都是基于声音的（6/6）。然而，根据方法 和打鼾（1/6）、发声（3/6）、打喷嚏（1/6）或啰音（1/6 能指标在表4中呈现。

。

特异性 (%)	精确度 (%)	准确度 (%)	F1分数 (%)
96.73	N/A	91.15	N/A
N/A	86.60	97.60	N/A
N/A	88.40	N/A	N/A
N/A	N/A	97.00	N/A
N/A	94.40	93.80	94.20
N/A	96.54	98.50	97.33

[45] 在田间条件下取得了良好的性能指标（灵敏度、精确度、准

an等人 [44,46] 开发了旨在检测和诊断家禽不同类型呼吸系 实验室条件下基于病毒接种前后发声开发的，并使用PCR测 [44] 在检测禽流感方面达到了高准确度，但未提供其他性能 度）。Banakar等人 [42]在检测禽流感方面取得了高性能 气管炎病毒和新城疫方面未能达到高灵敏度。Cuan等人 [4 得了高性能，并开发了一种旨在检测新城疫的技术。

的声音来监测呼吸系统疾病。Rizwan等人 [43] 监测啰音并 但无法实现高灵敏度（Se）和高精确度。Carpentier等人 声音的技术；然而，该技术的灵敏度（Se）较低，因此

ineffective to identify true positive cases under laboratory conditions.

3.4. Respiratory Disease PLF Technology

Several types of sensors were used in image-based (1/4), and accelerometers for bovine production are present.

Table 5. PLF performance measure

	Sensitivity (%)	Specificity
[47]	100.00	97.40
[13]	50.30	99.20
[48]	41.40	99.90
[16]	54.00	95.00

N/A—Not available.

Almost all technologies developed for diseases in dairy calves [13,16] and respiratory diseases in bovines.

The sound-based technologies; however, both technologies identified the disease.

Another study used an accelerometer to detect diseases in dairy calves [16]. This study achieved a high Sensitivity; however, and Carpentier et al. [48], the Sensitivity

3.5. Respiratory Disease PLF Technology

All studies assessed were technologies for swine production developed for the fattening stage; however, in this study no information was provided.

Table 6. PLF performance measure

	Sensitivity (%)	Specificity (%)	Precision (%)
[30]	N/A	N/A	N/A
[31]	N/A	N/A	N/A
[32]	N/A	N/A	N/A
[33]	N/A	N/A	N/A
[34]	N/A	N/A	N/A
[35]	N/A	N/A	N/A
[36]	N/A	N/A	N/A
[37]	92.00 ¹	N/A	90.80 ¹
[15]	98.60 ¹	N/A	95.50 ¹
[39]	97.72	95.01	96.81
[40]	96.80	93.20	95.50
[41]	96.51	N/A	98.41

¹ Performance measures of specific diseases.

Many technologies for disease detection in swine have been developed over the last two decades [30–33], and

Most technologies aimed at detecting diseases were not developed to detect a specific disease; however, they were mostly assessed by these studies of the technology in detecting

in identifying true positive cases in laboratory conditions.

3.4. Respiratory Disease PLF Technology

Several types of sensors were used in image-based (1/4) and accelerometers (1/4) for bovine production.

Table 5. PLF performance measure

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[47]	100.00	97.40
[13]	50.30	99.20
[48]	41.40	99.90
[16]	54.00	95.00

N/A—Not available.

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3.5. Respiratory Disease PLF Technology

All studies assessed were technologies for swine production developed for the fattening stage; however, in this study no information was provided.

Table 6. PLF performance measure

	Sensitivity (%)	Specificity (%)	Precision (%)
[30]	N/A	N/A	N/A
[31]	N/A	N/A	N/A
[32]	N/A	N/A	N/A
[33]	N/A	N/A	N/A
[34]	N/A	N/A	N/A
[35]	N/A	N/A	N/A
[36]	N/A	N/A	N/A
[37]	92.00 ¹	N/A	90.80 ¹
[15]	98.60 ¹	N/A	95.50 ¹
[39]	97.72	95.01	96.81
[40]	96.80	93.20	95.50
[41]	96.51	N/A	98.41

¹ Performance measures of specific diseases.

Many technologies for disease detection in swine have been developed over the last two decades [30–33], and

Most technologies aimed at detecting diseases were not developed to detect a specific disease; however, they were mostly assessed by these studies of the technology in detecting

sitive cases of the disease. Both studies were conducted unde

echнологies for Bovine Production

were used in bovine studies, sound-based technology (2/4
erometer (1/4). The performance measures of the technologie
esented in Table 5.

ures for bovine production.

city (%)	Precision (%)	Accuracy (%)	PPV (%)	NPV (%)
.40	N/A	N/A	86.30	100.00
.20	87.50	N/A	N/A	N/A
.90	94.20	N/A	N/A	N/A
.00	N/A	75.00	N/A	N/A

developed for bovine production aimed to detect respirator
[16,48]. This is in line with the period of highest risk fo
es [49].

ologies [13,48] were developed and validated in field cond
gies were inefficient to correctly identify true positive cases

accelerometer embedded in a necklace to detect respirator
. The accelerometer monitors feeding and activity behavior
; however, like the technologies from Vandermeulen et al. [1
e Se was low.

echнологies for Swine Production

ere sound-based (14/13). The performance measures of th
duction are presented in Table 6. Most technologies wer
stage (9/13), followed by finishing stage (3/13), and in on
ovided about the production stage.

ures for swine production.

Accuracy (%)	F1-Score (%)	Cough Detection Rate (%)
N/A	N/A	94.80
N/A	N/A	94.80
N/A	N/A	90.00
86.20	N/A	N/A
N/A	N/A	82.00
N/A	N/A	88.00
86.20	N/A	85.50
91.00 ¹	N/A	94.00
N/A	94.70 ¹	99.00
96.68	97.26	97.72
95.40	96.20	96.80
97.35	97.46	96.51

disease detection. N/A—Not available.

for swine production presented good performanc
e respiratory disease automatic detection has been researche
], enabling a myriad of studies on this topic.

d at detecting coughs to monitor respiratory health and thu
a specific respiratory disease [30–36,38–40]. The performanc
dies was the cough detection rate, which refers to the accurac
ng cough sounds. Many studies just presented this value t

效果不佳。这两项研究都是在实验室条件下进行的。

于牛生产

型的传感器，包括基于声音的技术（2/4）、基于图像的（
生产的PLF技术性能指标在表5中呈现。

(%)	精确度 (%)	准确度 (%)	阳性预测值 (%)	阴性预测值 (%)
.40	N/A	N/A	86.30	100.00
.20	87.50	N/A	N/A	N/A
.90	94.20	N/A	N/A	N/A
.00	N/A	75.00	N/A	N/A

技术都旨在检测奶牛小牛的呼吸系统疾病 [13,16,48]。这与
期 [49]一致。

是在实际田间条件下开发和验证的；然而，这两种技术都不
。

入的加速度计来检测奶牛小牛的呼吸系统疾病 [16]。加速度
研究实现了很高的特异度；然而，与Vandermeulen等人
的技术一样，灵敏度很低。

技术

声音的（14/13）。猪生产的技术的性能指标在表6中呈现。
的（9/13），其次是出栏阶段（3/13），在一项研究中没有

准确度 (%)	F1分数 (%)	咳嗽检测率 (%)
N/A	N/A	94.80
N/A	N/A	94.80
N/A	N/A	90.00
86.20	N/A	N/A
N/A	N/A	82.00
N/A	N/A	88.00
86.20	N/A	85.50
91.00 ¹	N/A	94.00
N/A	94.70 ¹	99.00
96.68	97.26	97.72
95.40	96.20	96.80
97.35	97.46	96.51

可用。

良好的性能指标 [15,37,39,40]。猪呼吸道疾病自动检测已研
该主题的研究众多。

测呼吸健康的技术并非为检测特定呼吸系统疾病而开发 [30–
的性能指标是咳嗽检测率，即该技术在检测咳嗽声音方面的
数值以

“prove” technology’s efficiency; could be considered reliable by

The technologies develop performance in the detection of recordings as the reference test,

Another set of studies from that aimed to identify specific s Postweaning Multisystemic Wa Syndrome virus) by monitoring performance > 90% for Se, precision, the disease, the reference test the disease. The cough detection reference test of Shen et al. [39] and video recordings and their footage that could help conclude

4. Discussion

We found promising results for respiratory diseases in livestock performance; however, when test and study condition), manual detection of livestock respiratory process in field conditions. We performed an initial analysis of technology to assess technology performance

Most of the technologies available. The most common respiratory disease and thus it is possible to monitor disease is more complex than of multiple signs (e.g., cough, fever). Articles excluded from this review quality in swine production [5] diseases, an alert based on air quality

Monitoring more than just create a technology that could monitor cough and fever together could monitor nose respiratory diseases. Other to overcome false positives, for particles in the air causing irritation. One article used a camera to monitor presented information that could be monitored for detecting respiratory for elevated body temperature. support decisions on the health monitoring specific clinical signs

Due to the current concern, diseased individuals could be whole group as only a single animal. poultry and swine attempted to study on dairy calves was able to use an accelerometer attached to the development of PLF technology individual using bioacoustic sensor through voice recognition could

“证明”技术的效率；因此，可靠性。

沈等人开发的 [39,41] 和尹这些研究将音频标注录音作为参

同一研究小组的另一个研究猪呼吸道疾病（支原体肺炎、断术。两项研究在灵敏度、精确度技术旨在检测疾病，因此参考研究的咳嗽检测率也达到了 >90% 频被标记为音频和视频录音，因此这可能导致更可靠的数据集

4. 讨论

我们发现，针对畜牧业中监测的结果。一些研究能够实现高条件）时，许多技术被认为不适行验证过程。我们强调，实验室下的测试是评估技术在真实场景

大多数用于呼吸道疾病检测导致动物发出声音 [9,14,50]，监测一个临床体征更为复杂，需变）。有两篇被排除在本综述之养猪生产中的空气质量 [51,52]。质量的警报并不一定意味着动物

监测呼吸道疾病的多种临床要；例如，同时监测咳嗽和发烧他未由该技术收集的数据也可以加空气中的细颗粒物，吸入后可摄像头监测动物体温 [47]。尽管吸道疾病可靠性的信息，但许多支持对动物健康状况做出决策很测目标疾病更受青睐 [20]。

由于当前对耐药性 [53] 的素的使用量，因为只需治疗单个别患病个体，只有一项关于奶牛连接到小牛 [16] 的加速度计。当个体的PLF技术以监测呼吸健康术，用于识别受影响的个体。

a disease. We understand that research to overcome usability

4.1. Performance Measures

Overall, Se was the most p that the algorithm must correct

Compared to Se, Sp was p technology to correctly identify low Sp could lead to an incre system's usage (e.g., farmer sto

In assessing the system's insights. Evaluating these perf alarm given by a technology i increase customer's confidence

The equation for Se is pres Technologies with low Se will because too many false negativ

Precision (i.e., number of t measure that can be useful for u for precision is presented as "Pr Sp is presented as "Sp = true n indicate the number of false pos or when clinical signs are abse or clinical sign), precision will positives (i.e., how many clinica technology). Although they are does not matter (e.g., when treat whether there is a percentage of signs), precision may be used i swine or poultry did not prese production, pigs are grown in g of a disease [55]. However, in bovine production) Sp should b

In this review, we consid presented by articles to suppo performance measure is more i Se has little impact and Sp is ver Sp instead of Se.

Analyzing performance m technology will be to monitor a all relevant performance measu support technologies' validity.

The nomenclature used by der interpretability. Exadaktylo all performance", which is the s way to overcome this would be t could help the reader to unders ideal since using many nomen to confusion. For example, the be the same as Se, since many a detect truthful cough sounds (tr negatives). To prevent mistaken measure with the nomenclature

一种疾病。我们理解，要开发这

4.1. 性能指标

总体而言，灵敏度是最常被个体的能力 [54]。

与灵敏度相比，特异度在更的能力 [54]。特异度低的技术可民停止关注警报）或抗生素过度

在评估系统有效性时，灵敏能指标可以展示技术发出的警报高客户对警报的信任度。

Se 的公式表示为 “Se = 真病动物或监测临床体征，因为该

精确度（即所有阳性警报中性能指标。精确度的公式表示为 Sp = 真阴性/真阴性 + 假阳性

（例如，无论动物中健康动物的时），精确度可以代替Sp使用而牛的所有研究都提供了Sp。对所有动物进行治疗。然而，在Sp。

在本综述中，我们认为文章来支持性能指标的展示。然而，如果在一个特定情况下，灵敏度度。

分析性能指标可以为读者提发现文章应展示所有相关的性能预测值和阴性预测值），以支持

不同研究对部分性能指标所 [34] 提出了一种名为“总体性 [37]相同。克服这一问题的方法现的内容。然而，这并非理想方淆。例如，性能指标“咳嗽检测有咳嗽声音（真阳性 + 和假阴性告结果，我们决定使用文章中使

at the development of such a solution would require much more study and technological challenges.

It presented performance measure; this test shows the capacity to correctly diagnose diseased individuals [54].

It is presented in fewer articles. Sp indicates the capacity of the technology to identify individuals that are not diseased [54]. A technology with a decreasing number of false alarms, which in turn may reduce the need to stop paying attention to the alerts) or antibiotic overuse.

The validity of Se and Sp may give different but complementary performance measures together demonstrate how reliable a technology is, so we hypothesize that high Se and Sp can potentially reduce the alarms given.

It is presented as “ $Se = \text{true positive} / (\text{true positive} + \text{false negative})$ ”. It indicates the ability of the technology to correctly identify diseased animals or monitor clinical signs that will be detected by the technology.

Precision (the number of true positives in all positive alarms) is another performance measure for understanding the validity of PLF technology. The equation for precision is “ $\text{Precision} = \text{true positive} / (\text{true positives} + \text{false positives})$ ”. While Sp indicates the ability of the technology to correctly identify healthy animals, how many were mistakenly detected as diseased animals. It indicates the number of false positives considering true negatives (i.e., of all healthy animals in the sample, how many were mistakenly detected as diseased animals). It is not the same, in cases where the number of true negatives is applied to the entire group of animals regardless of whether they are healthy or that do not present any clinical signs. Instead of Sp. This could justify why many technologies focus on precision, while for bovine all studies presented Sp. In swine group pens and all animals are treated when there is a case in production where the specific individual is treated (e.g., a pig with a disease) would be presented.

It is considered that just Se and Sp, or Se and precision, could be used to report performance. However, the decision of which is more important will vary according to the objectives of the user. In a very impactful in a specific situation, the user should focus on

These measures can give the reader full insight into how helpful a technology is for a disease. Therefore, we find that the articles should present performance measures (Se, Sp, precision, accuracy, F1-score, PPV and PNV) together.

By different studies for some performance measures may have been used. Los et al. [34] presented a performance measure named “overall accuracy” which is the same as “total accuracy” [36] and “overall accuracy” [37]. It is important to present the equation for the performance measure, which helps the reader understand exactly what is being presented. However, this is not an enclosure to describe the same thing could lead the reader to misunderstand the performance measure “Cough detection rate” is likely that many articles describe it as the ability that the technology had to detect true positives (true positives) out of all cough sounds (true positives + false positives). Since we are keenly reporting results, we decided to report the performance measure used by the article.

这样的解决方案需要大量研究以克服可用性和技术挑战。

被提及的性能指标；这项测试展示了算法必须具备正确诊断患病

更少的文章中被提及。特异度表明技术正确识别非患病个体可能导致误报数量增加，进而可能减少系统使用（例如，农场使用）。

敏感度和特异度可能提供不同但互补的见解。综合评估这些性能指标的可靠性，因此我们假设高敏感度和特异度可能潜在地提

真阳性/真阳性 + 假阴性”。Se 值较低的技术将无法识别患病个体，该技术会检测到过多的假阴性。

真阳性的数量）是另一个可以用于理解PLF技术有效性的指标。精度 = 真阳性/真阳性 + 假阳性”，而Sp表示为 “ $Sp = \text{true negative} / (\text{true negatives} + \text{false positives})$ ”。虽然它们不相同，但在真阴性数量无关紧要的情况下的比例或是否出现临床体征，都会对整个动物群体进行治疗。这可以解释为什么许多针对猪或家禽的技术没有提供Sp。在猪生产中，猪在群饲栏中饲养，当出现疾病案例时 [55] 在需要对个体进行治疗的生产中（例如牛生产），应提供

文章仅通过灵敏度（Se）和特异度（Sp），或灵敏度与精度，哪种性能指标更为重要，将根据用户的目标而有所不同。灵敏度影响较小而特异度影响很大，用户应关注特异度而非灵敏度。

提供全面了解一项技术对疾病监测的助益程度。因此，我们应关注性能指标（灵敏度、特异度、精确度、准确度、F1分数、阳性预测率）来评估技术的有效性。

所使用的命名法可能会影响可解释性。Exadaktylos等人提出的性能指标，它与“总准确度” [36] 和“总体准确度” [37] 是展示性能指标的公式，这有助于读者准确理解所呈方案，因为使用多种命名法来描述同一事物可能导致读者混淆。精度很可能与Se相同，因为许多文章将其描述为技术在所检测的真实咳嗽声音（真阳性）的能力。为避免错误使用命名法来报告性能指标。

A standard nomenclature researcher would use the same believe that it is challenging to The inclusion of the confusion most performance measures co NPV). The confusion matrix is a be used in future meta-studies

4.2. Reference Test

In this review, we decided t our knowledge, there is no perf monitor the clinical signs. We as consider to be reliable for valid

The reference test is esse technology that achieved a high is likely to perform badly when may provide incorrect insights, can negatively impact animals [

Vandermeulen et al. [13] a test to confirm if the propose monitoring coughing in dairy c test to determine if an animal health scoring criteria, a diagno signs caused by respiratory dise to correctly identify cases of b respiratory disease clinical sig dairy calves [13]. It is observed i of the technology. The technol had lower Se when compared chart [13]. If applied in field c would be less effective to detect

Schaefer et al. [47] combi affected by respiratory disease illness was considered as a res a clinical score higher than 3 (auscultation), and low levels o each of these clinical signs to de effective reference test in this s when interpreting these result by many diseases, so just asses diseases might not be ideal, e are susceptible to many other conditions similar to auctions environments (e.g., commercial

For bovine production, the ered a reliable reference test to d phy is considered accurate to de the efficiency of many clinical phy for diagnosing respiratory that spontaneous cough was th diseases; however, the perfor Sp = 85.7%).

The use of the Wisconsin sc of 74.1% [61]. This result show

为PLF验证研究制定标准命 同的术语。尽管使用标准术语是 具有挑战性。包含混淆矩阵是有 能指标都可以计算 (Se、Sp、 重要，可用于评估结果中的偏倚

4.2. 参考测试

在本综述中，我们决定使用 美的测试（金标准）可以用于识 参考测试，并说明了我们认为对

参考测试对支持技术性能至 在商业环境中很可能表现不佳。 错误决策，对动物产生负面影响

Vandermeulen等人 [13] 认所提出的技术能否通过监测奶 量作为参考测试，以确定动物是 康评分标准，这是一种包含呼吸 参考测试后，血液分析未能正确 床体征是验证奶牛小牛 [13] 技 使用血液分析 [48] 作为参考测 其灵敏度较低。如果在田间条件 阳性病例时将效果较差。

Schaefer等人 [47] 结合不 如果动物表现出至少三种临床体 温升高、临床评分高于3（中度 细胞增多。本研究展示了针对每 研究中最有效的参考测试（具有 体温升高是一种可能由多种疾病 吸系统疾病可能并非理想，特别 疾病 [57]。此外，该技术在类似 商业农场）时可能无法表现得那

对于牛生产，关于如何定义 胸部超声检查被认为能准确检测 床评分系统在诊断呼吸系统疾病 [59,60]。Lowie等人 [60] 发现 然而，该临床体征的表现相对较

Wisconsin scoring 系统的 特异度 (Sp) [61]。这一结果表

re for PLF validation studies could be a solution since ever e terms. Although using standard terms would be ideal, w to teach and require every researcher to use the same term n matrix is helpful and likely the best option as, if presente could be calculated (Se, Sp, accuracy, precision, F1-score, PP is also important to assess risk of bias in the results and coul es on this topic.

d to use “reference test” instead of “gold standard” because t perfect test (gold standard) to identify respiratory diseases or t assessed the reference test used by studies and stated what v alidating PLF technologies for monitoring respiratory disease sential for supporting the performance of a technology. gh performance in a study based on an unreliable reference te en applied in a commercial setting. Poorly tested technolog ts, consequently leading users to wrong decision-making th ls [56].

] and Carpentier et al. [48] used blood analysis as a referenc sed technology was able to detect respiratory diseases b y calves. Both used the number of neutrophils as the referenc l was sick. Vandermeulen et al. [13] also used the Wisconsi nostic tool that includes the clinical evaluation of six clinic iseases. After comparing reference tests, blood analysis faile bovine respiratory disease, and therefore assessing specifi signs is a better reference test to validate the technology fo d in the results that the reference test affected the performanc nology that just used blood analysis [48] as a reference te ed to the technology that used the Wisconsin health scorin d conditions, the technology with unreliable reference test tect true positive cases of diseased animals.

bined different clinical signs to determine if an animal wa ase. If the animal presented at least three clinical signs, th espiratory disease. The clinical signs were high temperatur 3 (moderate to severe nasal discharge, cough, and crepita s of white blood cells and neutrophilia. The comparison detect respiratory disease was presented. Fever was the mo s study (high Se and Sp). However, caution should be take ults. High temperature is a clinical sign that can be cause ssessing this clinical sign for exclusively detecting respirator , especially if applied to younger animals as these animal er diseases [57]. Furthermore, the technology was tested i ns and could not perform so well when applied in differe cial farm).

there is a wider discussion on defining what could be consi o detect respiratory diseases. The use of thoracic ultrasonogr detect respiratory diseases in bovine [58]. Studies comparin al signs and clinical score systems to thoracic ultrasonogr ry diseases were published [59,60]. Lowie et al. [60] foun the clinical sign that best indicated a calf to have respirator ormance of this clinical sign is relatively poor (Se = 37.4

scoring system was tested and achieved a Se of 62.4% and S ows that the scoring chart is also not a perfect measure to b

命名法可能是一个解决方案，因为每位研究人员都会使用相是理想的选择，但我们认为让每位研究人员都使用相同术语有帮助的，并且可能是最佳选项，因为如果呈现，大多数性、准确度、精确度、F1分数、PPV、NPV）。混淆矩阵也很倚风险，并可用于未来关于该主题的元研究。

用“参考测试”而非“金标准”，因为我们认为目前没有完识别呼吸道疾病或监测临床体征。我们评估了研究中使用的对验证精准畜牧业技术监测呼吸道疾病可靠的方面。

至关重要。基于不可靠参考测试的研究中表现优异的技术，。测试不充分的技术可能提供错误信息，进而导致用户做出响 [56]。

3] 和Carpentier等人 [48] 使用血液分析作为参考测试，以奶牛小牛咳嗽来检测呼吸道疾病。两者都使用中性粒细胞数是否患病。Vandermeulen等人 [13] 还使用了威斯康星健吸道疾病引起的六种临床体征临床评估的诊断工具。在比较确识别牛呼吸系统疾病病例，因此评估特定的呼吸道疾病临技术更好的参考测试。结果显示参考测试影响了技术性能。测试的技术，与使用威斯康星健康评分图表 [13] 的技术相件下应用，具有不可靠参考测试的技术在检测患病动物的真

不同的临床体征来判断动物是否受到呼吸系统疾病的影响。体征，该疾病就被视为呼吸系统疾病。这些临床体征包括体度至重度鼻分泌物、咳嗽和捻发音）、白细胞水平低和中性每种这些临床体征检测呼吸系统疾病的比较结果。发烧是本有高灵敏度和特异度）。然而，在解释这些结果时应谨慎。病引起的临床体征，因此仅通过评估这一体征来专门检测呼别是如果应用于幼年动物，因为这些动物容易感染多种其他似拍卖会的条件下进行了测试，在应用于不同环境（例如，那么好。

义可靠参考测试以检测呼吸系统疾病，存在更广泛的讨论。测牛的 [58] 呼吸系统疾病。有研究比较了多种临床体征和临病方面的效率与胸部超声检查的效率，这些研究已发表现，自发咳嗽是提示小牛患有呼吸系统疾病的最佳临床体征较差（灵敏度Se = 37.4%，特异度Sp = 85.7%）。

的使用经过了测试，达到了 62.4% 的灵敏度（Se）和 74.1% 的表明，评分图表也不是一个完美的衡量标准。

used as a reference test for detecting the criteria for diagnosing respiratory taken into consideration [59].

We categorized one type of assess clinical data other than cough from audio files that were pre-performed by a person that listed compared the number of cough observer that used the audio label an underestimation of up to 94 labelled the audio files. Therefore reference test. This may have various studies reviewed.

4.3. Risk of Bias

Several studies lacked key risk of bias. Within these studies information about the devices [necessary since different sensors could perform differently. A full the technology works and to prove the software.

A few studies did not show used to cross-reference their results of the context where the technology or animal production phase [37

Descriptive information may not work as well when applied (productions stage) or larger groups animals). For example, for swine than the finishing stage. A technology not work well when applied in in different stages to support the

One study [38] did not prove stalled. This is necessary because to different results. Providing data if the technology is useful, taking could work well if placed close

4.4. Can We Reliably Detect Lives

Based on our definition of achieved through a comparison (similar) conditions and having these criteria [15]. This study was and specific disease detection (respiratory disease, and post-weaning The technology is described as information to assess risk of bias labeling cough sounds from video a sound was correctly labeled) disease affecting the pigs. The to validate the algorithm for its performance for both cough sounds limitation of the study is the low

用作检测呼吸系统疾病的参考测试标准也应予以考虑 [59]。

我们将一种参考测试类型归总体而言，该方法包括对先前在一个人执行，该人聆听音频文件的咳嗽数量与使用先前描述的音频文件标记的观察者检测到的音可能并非有效的参考测试。这

4.3. 偏倚风险

一些研究缺乏被认为对评估问题与未提供有关所使用设备 [为不同的传感器（即不同的制造解技术的工作原理，并证明技术

一些研究未提供可用于交叉其他研究缺乏关于技术应用背景 [37]。

关于动物的描述性信息至共同生产阶段）或更大群体（将技术生产中，大多数技术是为育肥阶段若应用于不同的生产阶段，效果下的效率至关重要。

一项研究 [38] 未提供养殖在不同环境中应用可能导致不同非常重要，需要考虑其定位；例远时则可能无效。

4.4. 能否通过精准农业可靠地检

根据我们对可靠性的定义，在领域（或类似）条件下进行且 [15]。这项研究验证了一种能够肺炎、猪繁殖与呼吸综合征以及断成本解决方案，适用于小型农场种参考测试：从视频和音频记录以及血液分析以确定影响猪的呼特定疾病检测算法。该技术在咳的唯一局限性是样本量较小。

etecting respiratory disease. When applying scoring system respiratory diseases in different environments should also b

of reference test as audio labeled for those studies that did n cough. Overall, the methodology consists of labeling sound reviously recorded at animals' sites. The labeling process i stens to the audio files and label cough sounds. Aerts et al. [2 ughs that were labelled by an observer on the scene with a labelled methodology previously described; this study foun 94% on the number of coughs detected by the observer th efore, labeling sounds from recordings may not be an effectiv artificially inflated the performance measures reported in th

ey information that are considered important to assess studie dies, the most common issues were related to not presentin s [32,33,43], and software [47] used. Hardware information i ors (i.e., different manufacturers and technology specification full description of the software is needed to understand ho present proof that the technology's efficiency is supported b

ow the data (i.e., raw data or confusion matrix) that could b e results [37,41,44]. Other studies lacked descriptive informatio nology was applied such as the number of animals used [3 [37].

n regarding animals is essential since the technology coul ied to different groups (different ages, weights, and differe groups (applying the technology to a much larger group wine production most were developed for the fattening stag echnology developed and validated in a specific stage coul in a different production stage. It is important to conduct test t the technology's efficiency in different situations.

provide housing information and how the devices were i use technologies applied in different environments could lea data on how the sensor is installed is important to understan king in consideration its positioning; for example, a technolog se to animals but not work when placed more distantly.

vestock Respiratory Disease through Precision Farming?

of reliability, which included high technology performanc ison against reliable reference tests, conducted in field (o ing low risk of bias, we identified only one study that y validated a technology able to monitor pigs' cough sound n (*Mycoplasma hyopneumoniae*, porcine reproductive an weaning multisystemic wasting syndrome) in field condition as a low-cost solution that is suitable for smaller farms. A bias is described by the study. Two reference tests were use video and audio recordings (video was used to ensure th e), and blood analysis to determine the type of respirator he cough sound was linked to the disease affecting the pig r specific disease detection. The technology achieved hig h sound detection and specific disease detection. The onl low sample size.

测试。在应用评分系统时，不同环境下诊断呼吸系统疾病的

归类为音频标记，用于那些未评估咳嗽以外临床数据的研究在动物现场录制的音频文件中的声音进行标记。标记过程由件并标记咳嗽声音。Aerts 等人 [29]比较了现场观察者标记音频标记方法观察者标记的咳嗽数量；这项研究发现，使用的咳嗽数量存在高达 94% 的低估。因此，从录音中标记声这可能导致所审查研究中报告的性能指标被人为夸大。

估研究偏倚风险重要的关键信息。在这些研究中，最常见的 [32,33,43], 和软件 [47] 的信息有关。硬件信息是必要的，因造商和技术规格) 可能表现不同。需要完整描述软件，以了术的效率得到软件支持。

叉验证其结果的数据 (即原始数据或混淆矩阵) [37,41,44] 景的描述性信息，例如所用动物的数量 [32]或动物生产阶段

关重要，因为当技术应用于不同群体 (不同年龄、体重和不技术应用于更多动物) 时，其效果可能不佳。例如，在养猪阶段开发的，而非出栏阶段。在特定阶段开发并验证的技术果可能不佳。在不同阶段进行测试对于支持技术在各种情况

殖设施信息以及设备如何安装的细节。这是必要的，因为技同结果。提供传感器安装方式的数据对于理解技术是否有效例如，技术若放置在动物附近可能效果良好，但放置距离较

检测牲畜呼吸系统疾病?

，该定义包括通过与传统参考测试对比所实现的高技术性能且具有低偏倚风险，我们仅识别出一项符合这些标准的研究能够在领域条件下监测猪咳嗽声音和特定疾病检测 (支原体肺断奶后多系统消耗综合征) 的技术。该技术被描述为一种低场。该研究详细描述了所有评估偏倚风险的信息。使用了两录中标记咳嗽声音 (使用视频是为了确保声音被正确标记) 呼吸道疾病类型。咳嗽声音与影响猪的疾病相关联，以验证咳嗽声音检测和特定疾病检测方面均实现了高性能。该研究

Other two studies complied. One swine study [37] did not display their performance results. The thermography in bovine production.

A limitation of our approach >90% resulted in leaving several is that we have not assessed a internal validations available o

5. Conclusions

We evaluated PLF technology in poultry productions. One study was reliable. Some studies achieved would perform well when applied on laboratory conditions or use how studies report their validation measures, low performance measures within studies, and no

We encourage future studies so that all important information the effectiveness of a technology

The fact that some studies have been possible to rely on PLF technology

Supplementary Materials: The PRISMA flow diagram is available at <https://www.mdpi.com/article/10.3390/animals130921758543>.

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其他两项研究符合我们的大多数研究人群及其支持性能结果。然而，它未提及

我们方法的一个局限性可能未能进行进一步分析。另一个局限是精准畜牧业（PLF）会议记录

5. 结论

我们评估了精准畜牧业（PLF）技术在鸡生产中的应用。一些研究在实验室条件下或如何使用研究报告其验证措施，低性能措施，同一研究内对相同评估评估风险。

我们鼓励未来的研究改进方法，以全面了解技术的有效性。

一些研究表明其表现良好，呼吸道疾病的自动监测。

补充材料: 本稿件的PRISMA清单可在线查看，地址为：<https://www.mdpi.com/article/10.3390/animals130921758543>。

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lied with most of our criteria but had a high risk of bias [37,47] t describe the study population and the raw data supportin he other study monitored respiratory diseases with infrare uction [47]; however, it did not provide full software descript oach may have been that setting the performance threshold ral relevant articles out of further analysis. Another limitatio d all the PLF proceedings from conferences and companie e online.

ologies to monitor respiratory diseases in swine, bovine, an tudy met all the criteria set in this review to be considere ved high performance, but it is unclear if the technologie plied in real-life situations. Many studies were only conducte used an unreliable reference test. We identified issues wit idation tests, such as not reporting all relevant performanc measures, different nomenclature for the same performanc not reporting all necessary information to assess risk of bia dies to improve how the methodology and results are reporte ation is provided for readers to have a full understanding o logy.

s have had good performances indicate that, in the future, it mig nologies for automatic monitoring of respiratory diseases.

PRISMA checklist for this manuscript can be downloaded at: <http://3390/ani13071273/s1>, and Supplementary Table S1 at: <https://doi.org/10.3390/ani13071273/s1>, accessed on 14 March 2023.

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大部分标准，但存在高偏倚风险 [37,47]。一项猪研究 [37] 果的原始数据。另一项研究使用红外热成像法监测牛生产中 提供完整的软件描述

能是，将性能阈值设置为>90%导致排除了几篇相关的文章 局限性在于，我们没有评估所有会议和公司内部可在线获取 录。

PLF) 技术用于监测猪、牛和家禽生产中的呼吸道疾病。一 标准，被认为可靠。一些研究取得了高性能，但尚不清楚这 。许多研究仅在实验室条件下进行，或使用了不可靠的参考 验证测试时存在一些问题，例如未报告所有相关性能指标、 同性能指标使用不同的命名法，以及未报告所有必要信息以

方法学和结果报告，以便为读者提供所有重要信息，使其能

，这表明，在未来，我们可能能够依靠精准畜牧业技术进行呼吸

可从以下地址下载：

[/10.3390/ani13071273/s1](http://10.3390/ani13071273/s1)，以及补充表S1在：<https://doi.org/10.3390/ani13071273/s1>，数据获取日期为2023年3月14日。

、方法学、研究、撰写—初稿；S.T.M.S.: 方法学、研究、撰写— 写—审阅和编辑；R.R.D.: 概念提出、方法学、监督、项目行政管 者均已阅读并同意文稿的最终发表版本。

le.

或分析新数据。数据共享不适用于本文。

创新部及国家科学技术发展委员会（CNPq）通过项目编号

益冲突。

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